VETERINARY MICROBIOLOGY · RAPID REVISION

Unit 2 — Veterinary Mycology

All 10 syllabus topics on four working sheets. Fungi are a small unit that carries disproportionate marks because the questions repeat: name the fungus from its microscopic picture, name the toxin from the lesion, name the medium. This sheet set is built around exactly those three reflexes.

PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper I · Units 1, 2, 3 Theory 100 · Practical 60	10 of 10 covered	20 min full pass 8 min sheet 04 only	B.V.Sc & A.H. II Yr VCI MSVE · 3+2=5

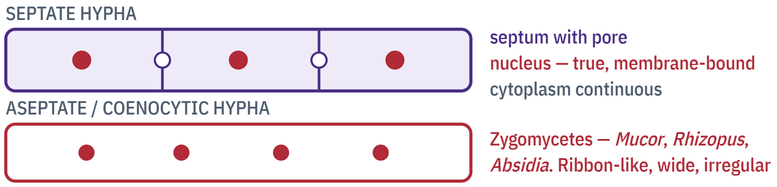
Sheet	What it covers	Syllabus topics folded in
01	The fungal cell, classification, growth & reproduction, lab toolkit	t01–t02
02	Yeasts & yeast-like fungi — *Candida*, *Cryptococcus*, *Malassezia*; dermatophytes	t03, t05
03	Moulds, dimorphic fungi, zygomycetes, mycetoma	t04, t06–t08
04	Mycotic mastitis & abortion, mycotoxicoses, antifungals	t09–t10

General mycology · Yeasts & dimorphic · Moulds · Dermatophytes · Mycotoxins · Exam trap & mnemonic

The Fungal Cell, Classification & Reproduction

What a fungus is — and is not

MECHANISM



Wall = CHITIN + glucan + mannan
(bacteria = peptidoglycan) → β-lactams useless
Membrane sterol = **ERGOSTEROL** (mammals = cholesterol)

Eukaryote:
true nucleus, 80S ribosome, mitochondria, ER, Golgi. Heterotroph, absorptive nutrition

Why this matters therapeutically: every antifungal in the exam targets one of the two things above — **ergosterol** (polyenes bind it, azoles block its synthesis, allylamines block squalene epoxidase) or **chitin/glucan** (echinocandins block β-1,3-glucan synthase). Griseofulvin is the exception: it binds tubulin and blocks mitosis.

Fungi vs bacteria

TABLE

Feature	Fungi	Bacteria
Cell type	Eukaryote	Prokaryote
Nucleus	True, membrane-bound	Nucleoid
Ribosome	80S	70S
Cell wall	**Chitin, glucan**	Peptidoglycan
Membrane sterol	**Ergosterol**	Absent (except *Mycoplasma*)
Size	2–10 µm and above	0.2–5 µm
Growth temp.	**25–30 °C** (room)	37 °C
Optimal pH	**5.6 (acidic)**	7.2–7.4
Growth speed	Days to weeks	Hours
Reproduction	Spores — sexual & asexual	Binary fission
Oxygen	Aerobe / facultative	Full range

SDA Sabouraud Dextrose Agar — 4% dextrose, peptone, **pH 5.6**, incubated at **25–27 °C**. Selectivity is added with **chloramphenicol** (kills bacteria) and **cycloheximide/actidione** (suppresses saprophytic moulds). **DTM** (dermatophyte test medium) adds phenol red → turns **red** with a dermatophyte.

Classification you must be able to write

TABLE

Division	Hyphae & sexual spore	Veterinary examples
Zygomycota	**Aseptate; zygospore**	*Mucor*, *Rhizopus*, *Absidia*, *Basidiobolus*, *Conidiobolus*
Ascomycota	Septate; **ascospore** in an ascus	*Aspergillus*, *Penicillium*, *Histoplasma*, *Blastomyces*, *Microsporum*, *Trichophyton*, *Candida* (teleomorph)
Basidiomycota	Septate + clamp connections; **basidiospore**	*Cryptococcus neoformans*, mushrooms, rusts & smuts
Deuteromycota "Fungi Imperfecti"	Septate; **no known sexual stage**	Older home of most veterinary pathogens; now redistributed by molecular data

- **Anamorph** = asexual (imperfect) state · **Teleomorph** = sexual (perfect) state · **Holomorph** = the whole fungus.
- **Morphological classes: Yeast** (unicellular, buds, pasty colony) · **Yeast-like** (buds + pseudohyphae — *Candida*) · **Mould/filamentous** (hyphae + mycelium, cottony colony) · **Dimorphic** (mould at 25 °C, yeast at 37 °C in tissue).
- **Clinical classification of mycoses: Superficial** (*Malassezia*, piedra) → **Cutaneous** (dermatophytes) → **Subcutaneous** (sporotrichosis, mycetoma, rhinosporidiosis) → **Systemic/deep** (histoplasmosis, blastomycosis, coccidioidomycosis) → **Opportunistic** (candidiasis, aspergillosis, cryptococcosis, zygomycosis).
- **Mycotoxicosis** and **mycetismus** (mushroom poisoning) are **intoxications, not infections** — no fungus grows in the animal.

Reproduction — spore names in one pass

RAPID-FIRE

Asexual — conidia	Borne on a conidiophore, exogenous, not enclosed
Microconidia / macroconidia	Small single-celled / large multiseptate — the basis of **dermatophyte identification**
Blastoconidia	Formed by **budding** — *Candida*, *Cryptococcus*
Arthroconidia	Hyphal fragmentation — *Coccidioides*, *Geotrichum*
Chlamydoconidia	Thick-walled resting cell; terminal chlamydospore on cornmeal-Tween 80 = ***C. albicans***
Sporangiospore	**Endogenous**, inside a sporangium — Zygomycetes
Phialoconidia	From a flask-shaped phialide — *Aspergillus*, *Penicillium*
Sexual — zygospore	Fusion of two similar gametangia (Zygomycota)
Ascospore	Usually 8, inside a sac (ascus)
Basidiospore	4, on the outside of a club-shaped basidium
Oospore	Fusion of unequal gametangia (Oomycetes — *Pythium insidiosum*)

The distinction that gets asked every year: sporangiospores are **endogenous** (formed inside a sac), conidia are **exogenous** (formed free at the tip). That one line separates Zygomycetes from everything else.

Growth requirements

RAPID-FIRE

Temperature	25–30 °C for moulds; 37 °C selects the **yeast phase** of dimorphics
pH	Acidic, ~5.6
Humidity	High; a_w > 0.70 permits mould growth on feed
Nutrition	Heterotrophic, absorptive; secrete exoenzymes
Time	Report a fungal culture as negative only after **3–4 weeks**
Oxygen	Aerobic; *Candida* facultatively anaerobic
Colony types	Glabrous (waxy), velvety, cottony/floccose, granular, powdery
Pigment	Note **obverse** and **reverse** colour — reverse pigment is a key ID clue

Diagnostic toolkit

RAPID-FIRE

10–20% KOH mount	Digests keratin, leaves fungal elements — skin, hair, nail
LPCB mount	**Lactophenol cotton blue**: lactic acid preserves, phenol kills, glycerol prevents drying, **cotton blue stains chitin**
India ink / nigrosin	Negative stain — ***Cryptococcus*** capsular halo in CSF
Slide culture	Riddell's technique — preserves conidial arrangement *in situ*
Germ tube test	Serum, 37 °C, 2–3 h — **Reynolds-Braude phenomenon, *C. albicans***
Special stains	**PAS** (magenta), **GMS** Grocott methenamine silver (black), calcofluor white + UV, mucicarmine for *Cryptococcus*
Wood's lamp	365 nm UV — **apple-green fluorescence, *Microsporum canis***
Serology / molecular	Latex agglutination for cryptococcal antigen, **galactomannan & β-D-glucan** for aspergillosis, AGID, ELISA, **ITS-region PCR & sequencing**, MALDI-TOF

Sheet 01 · 60-second self-test

RECALL

Prompt	Answer
Fungal cell wall polymer	Chitin (+ glucan, mannan)
Fungal membrane sterol	Ergosterol
pH and temperature of SDA	5.6 · 25–27 °C
Antibacterial added to SDA	Chloramphenicol
Inhibitor of saprophytic moulds	Cycloheximide (actidione)
Aseptate hyphae division	Zygomycota
Endogenous asexual spore	Sporangiospore
Dye in LPCB that stains chitin	Cotton blue
Slide culture technique	Riddell's
Germ tube phenomenon	Reynolds-Braude

02 Yeasts, Yeast-like Fungi & Dermatophytes

t03, t05

Candida albicans

SYSTEMATIC

- **Morphology:** Gram-positive oval budding yeast 2–6 µm; forms **blastoconidia, pseudohyphae and true hyphae** — the only common yeast that makes all three. Normal flora of gut, mouth, genital tract → a true **opportunist**.
- **Culture:** SDA 25–37 °C, 24–48 h — creamy-white, pasty, **yeasty smell**. On **CHROMagar**, *C. albicans* = green, *C. tropicalis* = blue, *C. krusei* = pink rough.
- **Two confirmatory tests: Germ tube test** — serum at 37 °C for 2–3 h gives a filament with **no constriction at its base** (constricted base = pseudohypha of another species). **Cornmeal-Tween 80 agar** at 25 °C → **thick-walled terminal chlamydospores**.
- **Predisposing causes:** prolonged broad-spectrum antibiotics, corticosteroids, immunosuppression, indwelling catheters, high-carbohydrate diet, moist skin folds.
- **Veterinary disease: crop mycosis / thrush (sour crop) in poultry** — thickened crop with a white pseudomembrane and "Turkish-towel" appearance; oral thrush and gastric candidiasis in young pigs and calves; mycotic mastitis in cows after intramammary infusion; vaginitis, cystitis and otitis in dogs.
- **Virulence:** adhesins, **phenotypic switching**, hyphal invasion, secreted aspartyl proteinases (SAPs), phospholipase, biofilm on catheters.
- **Treatment:** nystatin (topical/oral, not absorbed), fluconazole, amphotericin B; copper sulphate in poultry water; correct the predisposing cause.

Cryptococcus neoformans

SYSTEMATIC

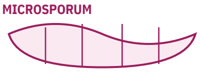
- **Morphology:** spherical budding yeast 4–20 µm with a **huge polysaccharide capsule (glucuronoxylomannan)** and a **narrow-based single bud**. **No pseudohyphae.**
- **Demonstration: India ink / nigrosin negative stain of CSF** — clear halo against a dark background; **mucicarmine** stains the capsule rose-red; latex agglutination for capsular antigen is the sensitive test.
- **Culture:** SDA **without cycloheximide** (it is inhibited by it) — mucoid cream colonies. **Urease positive**. **Birdseed / niger seed agar** → brown-black colonies from **phenol oxidase making melanin**.
- **Source: pigeon (and other avian) droppings** and soil; *C. gattii* from eucalyptus. Birds are carriers, not cases — their body temperature is too high.
- **Disease:** inhalation → **nasal granuloma with a "Roman nose" swelling in cats** (the classic case), sinusitis, pneumonia, **meningoencephalitis** with gelatinous cystic "soap-bubble" lesions in brain; mastitis in cattle; disseminated disease in dogs and horses. Zoonotically important in immunosuppressed people, but transmission is environmental, not animal-to-man.
- **Virulence:** capsule (antiphagocytic, the single most important factor), **melanin**, phospholipase, growth at 37 °C, urease.
- **Treatment:** amphotericin B + flucytosine; fluconazole penetrates CSF well; itraconazole for nasal forms.

> **Yeast in tissue — decide in one step:** capsule and narrow-based bud = *Cryptococcus* · pseudohyphae = *Candida* · **broad-based** bud = *Blastomyces* · **multiple buds, "mariner's wheel"** = *Paracoccidioides* · intracellular in macrophages = *Histoplasma* · spherule with endospores = *Coccidioides*.

Dermatophytes — identify by macroconidium

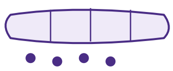
MECHANISM

MICROSPORUM



Large, spindle-shaped, **rough/echinulate** thick wall, 6–15 septa. Macroconidia abundant, microconidia few. *M. canis* — dog & cat, **Wood's lamp +ve**, reverse yellow. *M. gypseum* — geophilic, soil, horses; *M. nanum* — pigs.

TRICHOPHYTON



Cigar / pencil-shaped, **smooth thin** wall, 2–6 sep **Microconidia abundant** (grape-like clusters, abo *T. verrucosum* — **cattle ringworm**, needs thiamin grows best at 37 °C. *T. mentagrophytes* — rodent

EPIDERMOPHYTON — club-shaped smooth macroconidia in twos and threes, **NO microconidia**; skin &

Ringworm — the answer in eight lines

SYSTEMATIC

- **Definition:** dermatophytosis = keratinophilic infection of the **non-living keratinised tissue** — stratum corneum, hair and nail. The fungus never invades living tissue; the lesion is largely the host's inflammatory response.
- **Ecology: anthropophilic** (man) · **zoophilic** (animals — *M. canis*, *T. verrucosum*) · **geophilic** (soil — *M. gypseum*). **Zoophilic strains cause the most severe inflammation in man** — this is the standard zoonosis question.
- **Hair invasion pattern: Ectothrix** — arthroconidia sheathing the outside of the shaft (*Microsporum*, *T. verrucosum*) · **Endothrix** — inside the shaft, hair breaks at scalp level · **Favic** — hyphae + air spaces (*T. schoenleinii*, scutula).
- **Lesion:** circular, expanding, alopecic, scaly, grey-white crust with **central healing and an active periphery** — the "ring". In calves, classic on head, neck and around the eyes; in horses under the girth and saddle.

- **Diagnosis:** pluck hairs from the **edge** of the lesion → **KOH mount** for ecto/endothrix → **Wood's lamp** (only ~50% of *M. canis* fluoresce; *Trichophyton* does not) → SDA + chloramphenicol + cycloheximide at 25 °C for up to 4 weeks → LPCB for macroconidia → DTM turns red → PCR.
- **Treatment:** topical enilconazole, miconazole, 2% lime sulphur, povidone-iodine; systemic **griseofulvin** (with fatty food, **teratogenic — never in pregnancy**), itraconazole, terbinafine. Clip and burn hair, disinfect premises with hypochlorite; spores survive in the environment for years.
- **Prophylaxis:** live attenuated *T. verrucosum* vaccine (LTF-130) is used in cattle in some countries.

Malassezia pachydermatis

SYSTEMATIC

- **Morphology:** small oval-to-bottle-shaped budding yeast with a characteristic **broad-based "footprint" / peanut or bowling-pin shape** and a **monopolar unipolar bud with a collarette**.
- **Key point:** *M. pachydermatis* is **non-lipid-dependent** and therefore **grows on plain SDA** — unlike the human species (*M. furfur*) which need an olive-oil overlay.
- **Disease:** normal skin and ear-canal flora of dogs that overgrows with allergy, endocrinopathy, moisture or excess ceruminous secretion → **ceruminous otitis externa and seborrhoeic dermatitis of dogs**, greasy rancid smell, erythema and lichenification of skin folds. Also intertrigo and paronychia.
- **Diagnosis:** tape-strip or ear swab, stained with Diff-Quik — counting >5 yeasts per oil-immersion field supports the diagnosis. Culture is confirmatory.
- **Treatment:** miconazole-chlorhexidine shampoo, clotrimazole ear drops, oral ketoconazole/itraconazole, plus correction of the underlying allergy.

Other superficial **Piedra** — hair-shaft nodules: black piedra (*Piedraia hortae*, hard) and white piedra (*Trichosporon*, soft). **Pityriasis versicolor** — *M. furfur*, "spaghetti and meatballs" on KOH.

Sheet 02 · 60-second self-test

RECALL

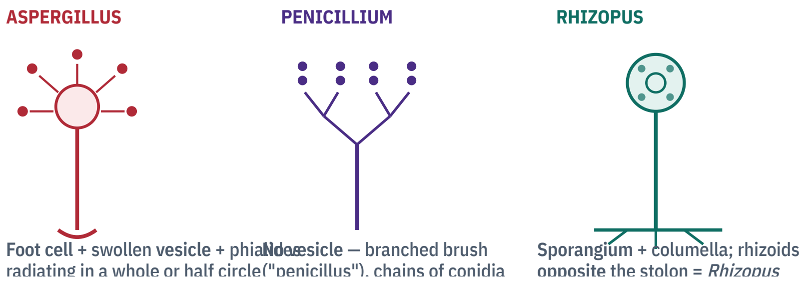
Prompt	Answer
Two confirmatory tests for *C. albicans*	Germ tube · chlamydospore on cornmeal agar
Sour crop / thrush in poultry	*Candida albicans*
Negative stain for *Cryptococcus*	India ink / nigrosin
Birdseed agar detects	Phenol oxidase → melanin, *Cryptococcus*
Environmental source of *Cryptococcus*	Pigeon droppings
Rough, spindle-shaped macroconidia	*Microsporum*
Smooth, cigar-shaped macroconidia	*Trichophyton*
Cattle ringworm agent	*T. verrucosum* (thiamine + inositol)
Wood's lamp positive dermatophyte	*M. canis* — apple-green
Antifungal contraindicated in pregnancy	Griseofulvin
Yeast of canine otitis externa	*Malassezia pachydermatis*

03 Moulds, Dimorphic Fungi & Subcutaneous Mycoses

t04, t06–t08

Aspergillus vs *Penicillium* vs Zygomycetes

MECHANISM



Zygomycete separation: *Rhizopus* — rhizoids **directly opposite** the sporangiophore · *Absidia* — rhizoids **between** two sporangiophores, pear-shaped sporangium with apophysis · *Mucor* — **no rhizoids at all**. All three are **angio-invasive**, causing thrombosis and infarction, and all grow fast, filling the plate in 2–3 days ("lid-lifters").

Aspergillosis & penicilliosis

SYSTEMATIC

- ***A. fumigatus*** — the chief pathogen; blue-green velvety colony, **uniseriate phialides on the upper half of the vesicle only, thermotolerant (grows at 45 °C)**. *A. flavus* — yellow-green, biseriate, radiate head, **aflatoxin**. *A. niger* — black, biseriate, radiate, ochratoxin, "black fungus" otomycosis. *A. terreus*, *A. nidulans* less common.
- **Brooder pneumonia of chicks** — the classic vet lesion: inhalation of spores from mouldy litter or hatchery contamination; **yellow-white caseous nodules/plaques in lung and air sacs**, gasping, high mortality in the first 2 weeks. Also **mycotic keratitis, encephalitis and ophthalmitis** in birds.
- **Other hosts: mycotic abortion in cattle** (raised, leathery placentitis with cotyledonary necrosis and ringworm-like plaques on the foetal skin); **guttural pouch mycosis** in horses (fatal epistaxis from internal carotid erosion); nasal aspergillosis in dogs (**turbinate destruction, *A. fumigatus***); air-sacculitis and disseminated disease in penguins and raptors.
- **Virulence:** small hydrophobic conidia (2–3 µm) reaching the alveoli, thermotolerance, gliotoxin, elastase, catalase, adhesion to fibrinogen and laminin.
- **Diagnosis: KOH/LPCB shows dichotomously branching septate hyphae at 45°**; culture on SDA **without cycloheximide;** histopathology with GMS/PAS; **galactomannan and β-D-glucan** assays; AGID; PCR. **Positive culture alone is not proof** — *Aspergillus* is a ubiquitous contaminant; correlate with lesions.
- ***Penicillium*** — mostly a contaminant and a toxin producer (ochratoxin A, citrinin, patulin, PR toxin); *Talaromyces (Penicillium) marneffei* is the one dimorphic species, causing disseminated disease in bamboo rats and man in South-East Asia.

Dimorphic fungi — the "25 °C / 37 °C" table

TABLE

Fungus	Mould form, 25 °C	Tissue/yeast form, 37 °C	Disease
Histoplasma capsulatum	**Tuberculate macroconidia** (thick-walled with finger-like projections) + smooth microconidia	Small **intracellular yeast inside macrophages,** 2–4 µm	Histoplasmosis — reticulo-endothelial, chronic wasting & diarrhoea in dogs; bird & bat guano source. **var. *farciminosum* = epizootic lymphangitis of horses** ("Japanese farcy") — corded lymphatics with ulcerating nodules
Blastomyces dermatitidis	Small round-to-pear-shaped conidia on short stalks ("lollipops")	Large 8–15 µm yeast, thick **doubly-contoured wall, single BROAD-BASED bud**	Blastomycosis — pyogranulomatous pneumonia, skin nodules, uveitis, bone lesions in dogs; river valleys, moist soil
Coccidioides immitis	**Barrel-shaped arthroconidia** alternating with empty cells — **highly infectious, BSL-3**	**Spherule (sporangium) 20–200 µm packed with endospores** — not a yeast	Coccidioidomycosis / "valley fever" — desert soil, dust-borne; granulomatous pneumonia and disseminated bone disease in dogs and horses
Paracoccidioides brasiliensis	Slow-growing, sparse conidia	Yeast with **multiple narrow-necked buds — "mariner's wheel / ship's wheel"**	South American blastomycosis; man; armadillo reservoir

Dimorphism Mould in nature and on SDA at 25 °C; **yeast (or spherule) in the host at 37 °C.** The switch is driven by **temperature** (and CO_2/cysteine for *Histoplasma*), and the **mould form is the infectious form** while the **yeast form is the invasive form** — that sentence answers the standard "define dimorphism with examples" question.

Subcutaneous mycoses

SYSTEMATIC

- ***Sporothrix schenckii* — sporotrichosis.** Dimorphic; 25 °C mould with **daisy/rosette clusters of conidia on delicate hyphae**; 37 °C **cigar-shaped yeast**. Entry by thorn or splinter → **"rose-gardener's disease"**; in horses and cats **ascending chains of nodules and ulcers along the lymphatics** (lymphocutaneous form). **Cats shed enormous numbers of organisms — a genuine zoonotic risk to handlers.** Asteroid bodies in tissue. Treat with itraconazole or oral potassium iodide.
- ***Rhinosporidium seeberi* — rhinosporidiosis. Cannot be cultured**; now placed with the aquatic protistan class Mesomycetozoea, not true fungi. Associated with stagnant pond water and river bathing. **Pedunculated, friable, red polyp studded with white spots ("strawberry polyp")** in the nose of cattle, horses, dogs and man; also conjunctiva. Histology shows **huge sporangia (up to 300 µm) full of endospores**. Treatment is **surgical excision with cautery of the base** — recurrence is common; dapsone is adjunctive.
- **Mycetoma (Madura foot).** A **triad**: tumefaction (swelling) + draining sinuses + **grains/granules** in the discharge. **Eumycotic** mycetoma = true fungi (*Madurella*, *Pseudallescheria boydii*, *Curvularia*) with **broad septate hyphae** in the grain; **Actinomycotic** mycetoma = filamentous bacteria (*Nocardia*, *Actinomadura*, *Streptomyces*) with **thin 1 µm filaments**. Grain colour and the Gram/ZN reaction of the grain decide which — and this decides the treatment: antifungals versus sulphonamides.
- **Zygomycosis (mucormycosis).** Opportunistic, in diabetic, acidotic, immunosuppressed or heavily antibiotic-treated animals. **Broad, aseptate, ribbon-like hyphae branching at wide (~90°) angles**, strongly **angio-invasive** → thrombosis, infarction and black necrotic tissue. Rhino-cerebral, gastric (calves, pigs) and abortion forms. Contrast with *Aspergillus*: septate, narrow, 45° branching.
- **Also name: pythiosis** ("swamp cancer" — *Pythium insidiosum*, an oomycete with kunkers, in horses and dogs) and **phaeohyphomycosis** (dematiaceous, melanised hyphae).

Sheet 03 · 60-second self-test

RECALL

Foot cell + vesicle	*Aspergillus*	Brush-like conidiophore, no vesicle	*Penicillium*
Brooder pneumonia	*A. fumigatus*	Guttural pouch mycosis	*A. fumigatus*
Rhizoids opposite the stolon	*Rhizopus*	No rhizoids	*Mucor*
Septate, 45° dichotomous branching	*Aspergillus*	Aseptate, 90° ribbon-like	Zygomycetes
Tuberculate macroconidia	*Histoplasma capsulatum*	Epizootic lymphangitis	*H. capsulatum* var. *farciminosum*
Broad-based single bud	*Blastomyces dermatitidis*	Spherule with endospores	*Coccidioides immitis*
Mariner's wheel	*Paracoccidioides brasiliensis*	Rosette/daisy conidia, cigar yeast	*Sporothrix schenckii*
Strawberry nasal polyp, non-cultivable	*Rhinosporidium seeberi*	Triad of swelling, sinuses, grains	Mycetoma

04 Mycotic Mastitis & Abortion · Mycotoxicoses · Antifungals

t09–t10

Mycotic mastitis

SYSTEMATIC

- **Agents: *Candida* spp. (commonest), *Cryptococcus neoformans*, *Trichosporon*, *Aspergillus fumigatus*, *Geotrichum*, *Prototheca*** (an achlorophyllous alga — always mentioned with this group).
- **How it happens — this is the whole answer:** it is nearly always **iatrogenic**. Repeated intramammary infusion of antibiotics with a contaminated or

Mycotic abortion

SYSTEMATIC

- **Agents: *Aspergillus fumigatus* accounts for 60–80%**; also *Mucor*, *Rhizopus*, *Absidia*, *Candida*, *Mortierella wolfii* (New Zealand, with a fatal maternal pneumonia).
- **Pathogenesis:** inhalation or ingestion of spores from **mouldy hay, silage or damp bedding** in winter-housed cattle → **haematogenous spread** →

multi-dose tube, unhygienic teat canal, and the suppression of normal bacterial flora by prolonged antibiotic therapy together allow fungal overgrowth.

- **Clinical:** swollen, firm, painful quarter with fever in the acute phase, then a **self-limiting or chronic granulomatous mastitis that does not respond to antibiotics** — often the first clue. Milk is watery with flakes; marked drop in yield; the quarter may become fibrotic.
- **Diagnosis:** aseptic milk sample → direct smear (Gram/LPCB/Giemsa) showing budding yeasts or hyphae → SDA + chloramphenicol at 25 and 37 °C → germ tube/CHROMagar → confirm by repeat sampling, since a single positive can be contamination.
- **Management: stop all antibiotic infusions immediately**, frequent stripping, supportive therapy, nystatin or clotrimazole infusion, iodine teat dip, cull chronic cases. Control = strict aseptic infusion technique and single-dose tubes.

> ***Prototheca zopfii*** is not a fungus but an **achlorophyllous alga**; it forms **morula-like sporangia with endospores**, grows on SDA, causes an incurable granulomatous mastitis (cull) and enteritis/disseminated disease in dogs.

localisation in the placentome → placentitis and foetal death. Not contagious cow-to-cow.

- **When:** sporadic, usually in the **last trimester (5–8 months)** in cattle; the dam is otherwise healthy.
- **Lesions — the two things to write: (1) Placenta** — severely thickened, leathery, necrotic cotyledons with a yellow-brown adherent exudate and oedematous intercotyledonary areas; retained. **(2) Foetus** — ~25% show **raised, grey-white, circumscribed "ringworm-like" plaques of dermatomycosis on the skin**, especially head, neck and shoulders — pathognomonic when present.
- **Diagnosis:** KOH/LPCB and GMS/PAS of cotyledon and foetal abomasal content showing **septate branching hyphae**; culture on SDA; histopathology showing hyphae invading placental vessels; **rule out *Brucella*, *Leptospira*, *Campylobacter*, IBR and BVD first** — say this in the answer.
- **Control:** no treatment. Prevent by avoiding mouldy feed, proper hay curing and silage compaction, ventilation, and mould inhibitors (propionic acid) in stored feed.

Mycotoxins — the single highest-yield table in this unit

TABLE

Mycotoxin	Producer	Substrate	Target organ & mechanism	Signs & the fact that gets asked
Aflatoxin B_1, B_2, G_1, G_2	*Aspergillus flavus*, *A. parasiticus*	Groundnut cake, maize, cottonseed	**Liver** — bioactivated by cytochrome P450 to the 8,9-epoxide which binds DNA	Most potent natural **hepatocarcinogen**. Ducklings & young poultry most susceptible; icterus, ascites, hepatic fibrosis, ↓growth, immunosuppression. **B_1 is excreted in milk as M_1** — the public-health point. Blue-green fluorescence at 365 nm; detected by TLC/HPLC/ELISA. **Turkey X disease, 1960**
Ochratoxin A	*A. ochraceus*, *Penicillium verrucosum*	Stored cereals, barley	**Kidney** — inhibits phenylalanine-tRNA synthetase	Nephropathy of pigs and poultry; pale swollen kidneys, polyuria, poor shell quality; also immunosuppressive and teratogenic
Fumonisin B_1	*Fusarium verticillioides* (*moniliforme*)	**Maize**	Blocks **ceramide synthase** → sphingolipid disruption	**Equine leucoencephalomalacia (ELEM) — "mouldy corn poisoning", liquefactive necrosis of cerebral white matter**; **porcine pulmonary oedema**; hepatotoxic in other species
T-2 toxin & DON (vomitoxin)	*Fusarium sporotrichioides*, *F. graminearum*	Wheat, maize, damp grain	**Trichothecene** — inhibits protein synthesis at the ribosome; radiomimetic	**Oral and gastric ulceration, beak & mouth lesions in poultry**, feed refusal and vomiting in pigs (DON), bone-marrow suppression, haemorrhagic syndrome, contact dermatitis; **alimentary toxic aleukia** in man
Zearalenone (F-2)	*Fusarium graminearum*	Maize, barley	**Oestrogen-receptor agonist**	**Hyperoestrogenism** — vulvovaginitis, swollen vulva and prolapse in gilts, mammary enlargement, splay leg, infertility, anoestrus; the classic **pig** mycotoxicosis
Ergot alkaloids	*Claviceps purpurea*, *C. fusiformis*	Rye, bajra, grasses	α-adrenergic vasoconstriction; dopamine agonism → ↓prolactin	**Gangrenous ergotism** — dry gangrene of extremities, ear tips, tail, hooves; convulsive form; agalactia in mares and sows. **Fescue toxicosis** (*Neotyphodium coenophialum* ergovaline) — fescue foot, summer slump, fat necrosis
Slaframine	*Rhizoctonia leguminicola*	Red clover ("black patch")	Cholinergic (muscarinic) agonist	**Slobbers syndrome** — profuse salivation, lacrimation, bloat
Sporidesmin	*Pithomyces chartarum*	Pasture litter (rye grass)	Bile-duct necrosis → phylloerythrin retention	**Facial eczema** of sheep — hepatogenous **secondary photosensitisation** of unpigmented skin
Citrinin · Patulin · PR toxin	*Penicillium* spp.	Rice, apples, silage	Nephrotoxic / cytotoxic	Yellow rice disease; often asked as a one-line item only

> **Diagnosis & control of any mycotoxicosis** **Diagnosis:** herd history of a new feed batch + multiple animals affected + no fever + no response to antibiotics + non-contagious → examine feed for mould, UV screening, TLC/HPLC/LC-MS/MS, ELISA kits, and a **feeding trial**. **Control:** harvest and dry to **<14% moisture (a_w < 0.70)**, aerated storage, mould inhibitors (propionic/sorbic acid), **adsorbents — hydrated sodium calcium aluminosilicate (HSCAS), bentonite, esterified glucomannan**, ammoniation for aflatoxin, dilution with clean feed, added antioxidants, methionine and vitamins. There is **no specific antidote** — remove the feed.

Antifungal drugs by target

TABLE

Class	Mechanism	Drugs & use
Polyenes	**Bind ergosterol** → membrane pores → leakage	**Amphotericin B** (systemic, nephrotoxic), **nystatin** (topical/oral only, not absorbed — candidiasis), natamycin
Azoles	Inhibit **14-α-demethylase (CYP51)** → blocks ergosterol synthesis	Imidazoles — miconazole, ketoconazole, clotrimazole, enilconazole; Triazoles — **fluconazole (best CSF penetration), itraconazole, voriconazole**
Allylamines	Inhibit **squalene epoxidase**	**Terbinafine** — dermatophytosis
Echinocandins	Inhibit **β-1,3-glucan synthase** (cell wall)	Caspofungin, micafungin — invasive candidiasis
Antimetabolite	Converted to 5-FU → blocks DNA/RNA	**Flucytosine** — always with amphotericin B for cryptococcosis
Others	Binds **tubulin**, blocks mitosis	**Griseofulvin** — dermatophytes only, **teratogenic**; potassium iodide for sporotrichosis; copper sulphate & gentian violet as feed/water antifungals

Antifungal sensitivity testing CLSI broth microdilution (M27 for yeasts, M38 for moulds) giving an **MIC**; disc diffusion and **E-test** gradient strips are the practical lab versions. Less standardised than bacterial ABST — say so.

Sheet 04 · 60-second self-test

RECALL

Prompt	Answer
Commonest cause of mycotic mastitis	*Candida* spp.
Alga causing bovine mastitis	*Prototheca zopfii*
Commonest cause of mycotic abortion	*Aspergillus fumigatus*
Skin plaques on aborted foetus	Foetal dermatomycosis
Most potent natural hepatocarcinogen	Aflatoxin B_1
Aflatoxin metabolite in milk	Aflatoxin M_1
Turkey X disease, 1960	Aflatoxicosis
Equine leucoencephalomalacia toxin	Fumonisin B_1
Hyperoestrogenism in gilts	Zearalenone
Vomitoxin	Deoxynivalenol (DON)
Facial eczema of sheep	Sporidesmin (*Pithomyces*)
Slobbers syndrome	Slaframine
Safe feed moisture level	Below 14%
Aflatoxin binder in feed	HSCAS / bentonite
Antifungal that binds ergosterol	Amphotericin B (polyene)
Antifungal binding tubulin	Griseofulvin

`10 / 10 syllabus topics` `Paper I · Unit 2` `Full pass ≈ 20 min`

`Night-before: sheet 03 diagrams + all self-tests`